Introduction in Machine Learning

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What is Machine Learning?

- A subset of Artificial Intelligence (AI)
- Enables systems to **learn patterns from data** and make predictions
- No explicit programming for each task
- **Goal:** Improve performance on a task using data



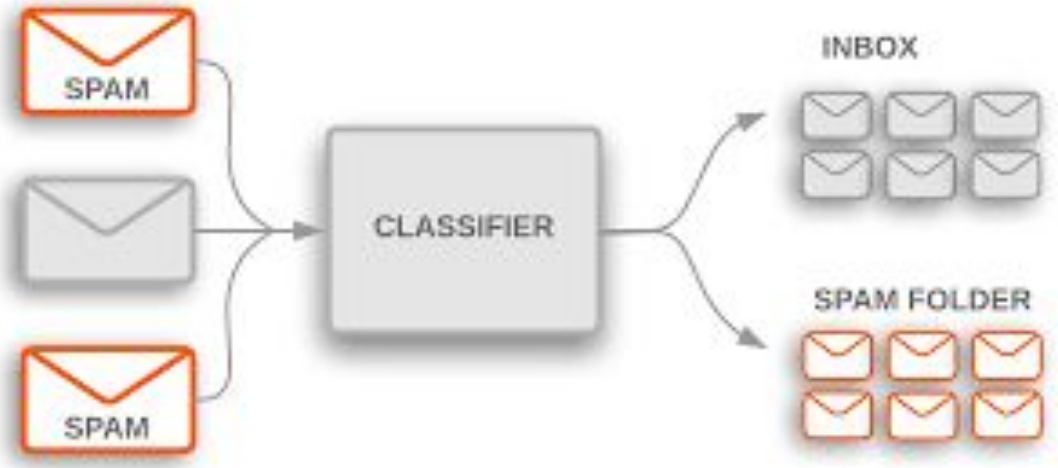
Types of Machine Learning -

Type	Description	Example Use Case
Supervised	Learn from labeled data	Spam detection, stock price prediction
Unsupervised	Learn from unlabeled data	Customer segmentation, topic modeling
Reinforcement	Learn via rewards and penalties	Robotics, Game playing

Supervised Learning

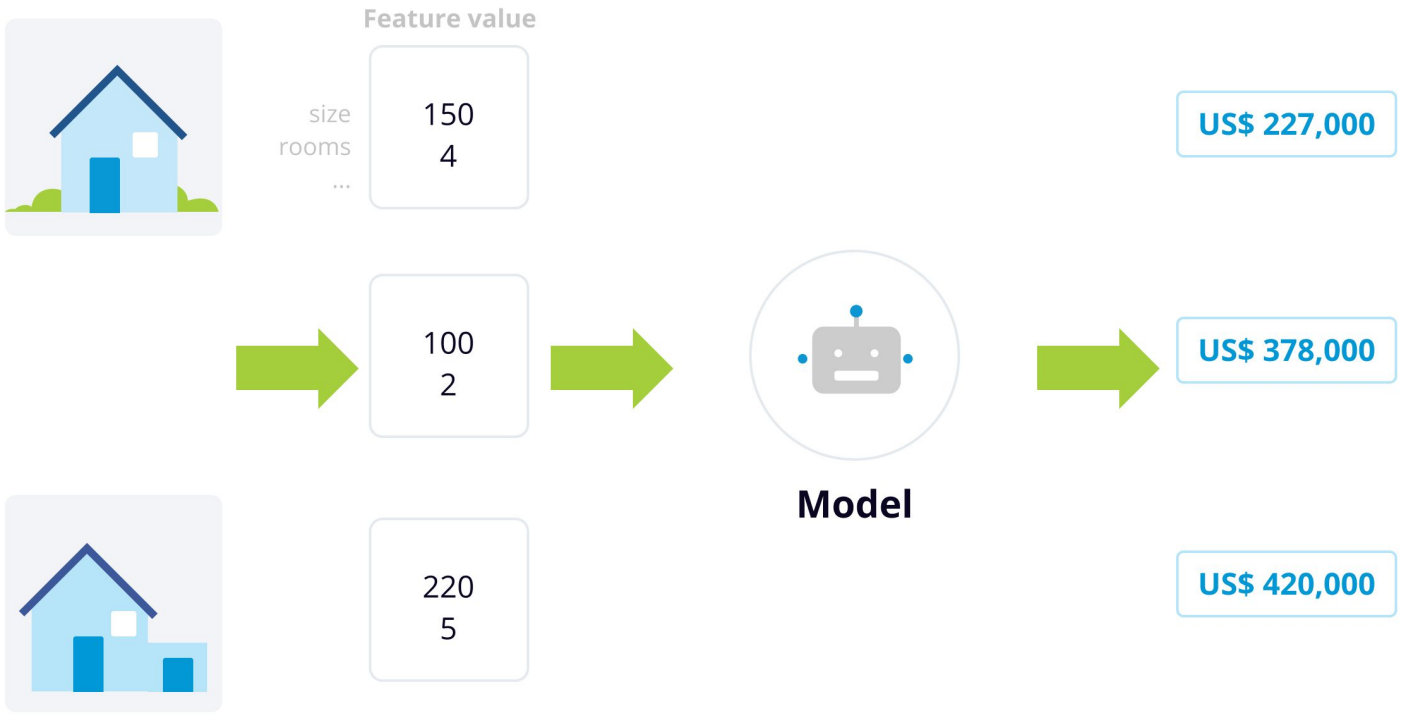
- Input data with **known labels**
- Algorithm learns to **map inputs to outputs.**
- **Two types:**
  - **Classification** – output is a category
  - **Regression** – output is a continuous value

- Examples:
  - Predicting whether an email is spam (Classification)
  - Predicting house prices (Regression)



Spam detection

House Pricing



Unsupervised Learning

- Input data **without labels**
- Algorithm identifies **hidden patterns** or structures
- Common techniques:
  - **Clustering** – group similar items (e.g. K-Means)

- **Example:**
  Grouping customers based on purchasing behavior

Classification vs. Regression

Task	Output Type	Example
Classification	Discrete category	Fraud detection (Yes/No)
Regression	Continuous value	Predicting sales ($)

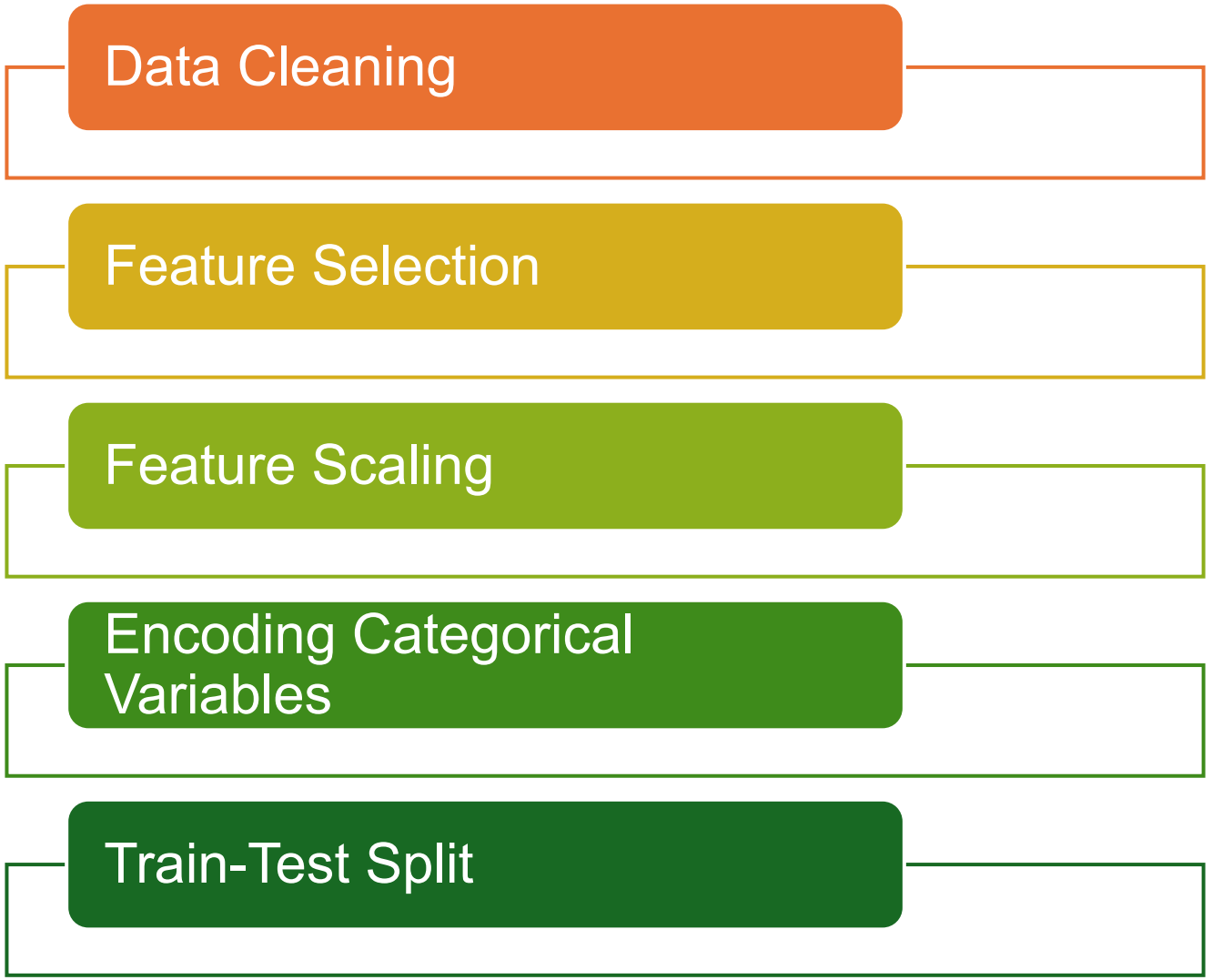
Importance of Preprocessing

Raw data is often **noisy**, **incomplete**, or **inconsistent.**

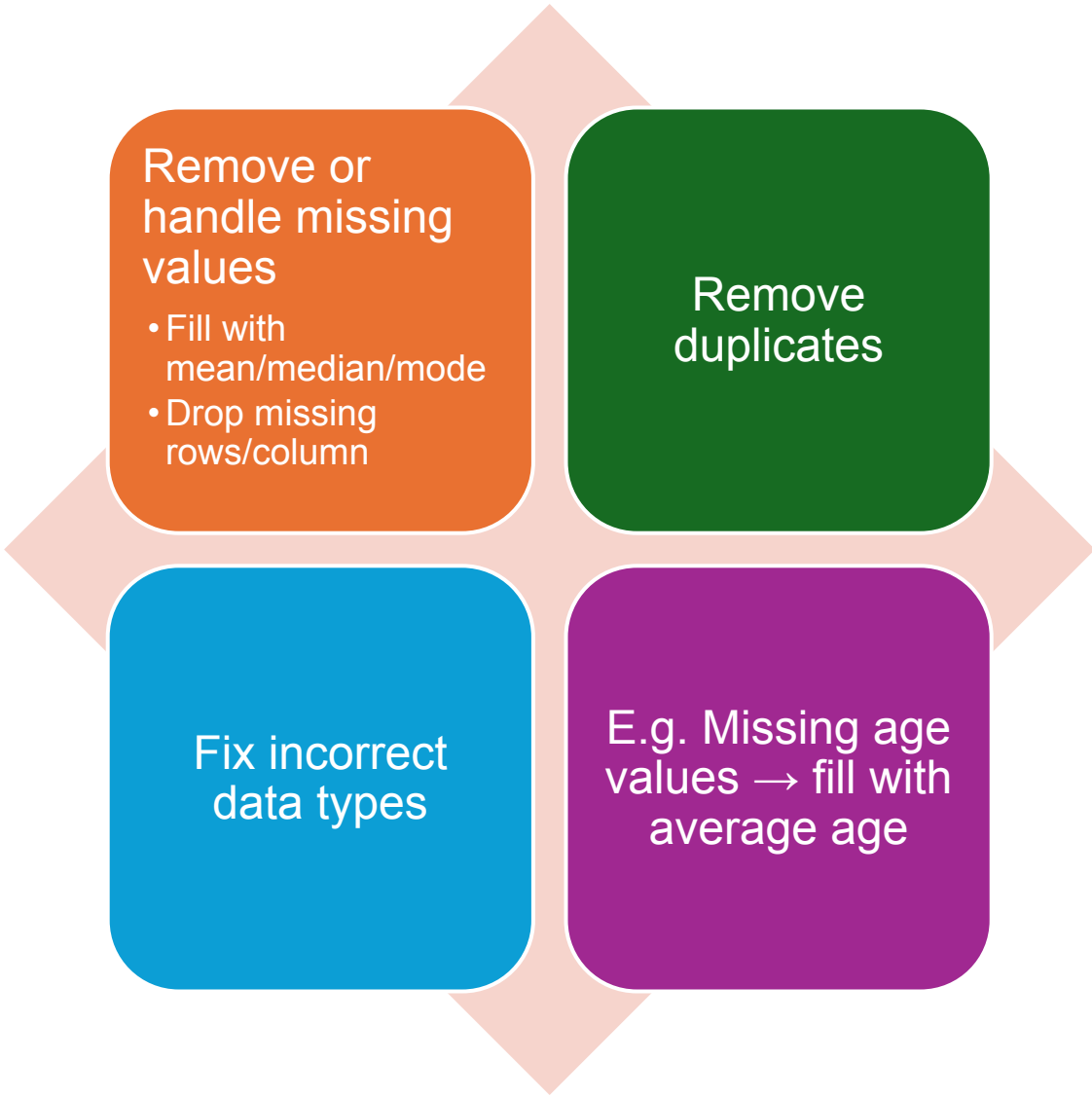
Preprocessing transforms raw data into a clean and usable format.

Improves model accuracy and efficiency

Common Preprocessing Steps



Data Cleaning



Prepocessing Cont'd

Feature	**Feature Selection** • Choose only relevant features for training
Scaling	**Feature Scaling** • Machine learning algorithms often require normalized or standardized data

Encoding Categorical Variables

- Convert non-numeric (categorical) data to numeric
- Techniques:
  - **Label Encoding** (e.g., Male → 0, Female → 1)
  - **One-Hot Encoding** (e.g., Red → [1,0,0], Blue → [0,1,0])

Train-Test Split

- Divide data into:
  - **Training Set**: Used to train the model
  - **Testing Set**: Used to evaluate performance
- Typical ratio: **70% train, 30% test** (or 80/20)

Example Workflow

- Load raw data
- Clean missing or bad values
- Encode categorical variables
- Scale numerical features
- Select important features
- Split into training/testing sets
- Feed into model (e.g., SVM or KNN)

